

FieldCommand Incident Management System

Open-Source · Offline-First · Field-Deployable

FIELDCOMMAND

Incident Management System v1.0

COMPLETE USER MANUAL

v1.0

ICS/NIMS All-Hazards · Amateur Radio EMCOMM · Public Safety

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1 Introduction & System Overview

When a disaster takes down commercial infrastructure — cellular networks, internet, power — the tools modern emergency management depends on disappear precisely when they are needed most. Incident logs revert to paper. Resource tracking becomes a whiteboard. ICS forms are filled out by hand, photocopied, and hand-carried between rooms. Situational awareness degrades to whatever one person can hold in their head. Every organization that has worked a major activation knows this failure mode, and most have simply accepted it as the cost of doing business.

FieldCommand IMS was built to eliminate that failure mode. It is a complete, self-contained incident management platform that carries its own network, its own server, and its own tools — and it operates with no internet connection, no cellular service, and no outside infrastructure of any kind. The cell towers are down, the internet is gone, and you are running on a generator in a parking lot. That is exactly when you need incident management software most — and that is exactly when FieldCommand IMS is designed to perform.

1.1 What FieldCommand IMS Is

FieldCommand IMS is a **complete ICS/NIMS all-hazards incident management system** — not simply an amateur radio or EMCOMM tool. It manages the full lifecycle of any incident from initial response through demobilization using standard ICS forms and workflows. It runs on a Raspberry Pi 5 server and broadcasts its own private Wi-Fi access point. Any smartphone, tablet, or laptop that joins that network immediately has access to the full suite of 48 tools through a standard web browser — no app installation, no accounts, and no per-device configuration required.

When internet connectivity is available, live features activate automatically: NWS weather alerts, APRS-IS, animated NEXRAD radar, and HF propagation data. If connectivity is lost at any point, all core tools continue without interruption. The system degrades gracefully and recovers automatically.

1.2 How FieldCommand IMS Differs From Other ICS Platforms

Platforms such as WebEOC, E-Team, NIMSAP, NIMS Logic, and E-iSuite deliver powerful incident management capability — but every one of them requires a working internet connection and functioning server infrastructure. They are cloud-dependent by design. When a major disaster disables that infrastructure, they fail with it — at exactly the moment capability is most critical.

FieldCommand IMS is built on the opposite assumption: infrastructure will fail, and incident management capability must survive that failure. It carries its own infrastructure — server, network, storage, and tools — in a single deployable package. It requires no IT department, carries no licensing fees, and does not fail when the internet fails.

The feature that sets FieldCommand IMS apart from every platform in this category is its native integration of amateur radio and public safety communications directly into the incident management workflow. No other ICS platform includes built-in net control logging, FCC callsign validation against the full national licensee database, APRS tactical mapping, Winlink radio email, JS8Call HF messaging, or AMPRNet gateway capability. These are not add-ons — they are core features fully integrated with the ICS platform so that radio traffic, net logs, and check-in data flow directly into ICS-309 communications logs, ICS-214 activity logs, and the IAP.

PLATFORM	OFFLINE?	SELF-CONTAINED ?	NATIVE EMCOMM?	COST
FieldCommand IMS	✓ Fully offline	✓ Own network + server	✓ Full EMCOMM suite	Free / open source
WebEOC	✗ Cloud only	✗ Requires infrastructure	✗ None	\$10,000–50,000/yr
E-Team / Veoci	✗ Cloud only	✗ Requires infrastructure	✗ None	Subscription
NIMSIAP	✗ Internet required	✗ Requires connectivity	✗ None	Free
E-iSuite	■ Limited	■ Laptop only, no network	✗ None	License fee

1.3 System Architecture

COMPONENT	FUNCTION	DEFAULT IP
FieldCommand Pi 5 16GB	Primary server · all 48 web tools · 12+ background services · RAID-1 NVMe storage · EMCOMM-NET Wi-Fi AP	192.168.50.1
44Net Gateway Pi 5 (optional)	AMPRNet WireGuard tunnel · callsign-authenticated access · Part 97 access log · isolated from primary server	192.168.50.2
Wi-Fi router (primary)	DHCP · AiMesh controller · dual WAN management · EMCOMM-NET SSID broadcast. Recommended: ASUS RT-BE58 Go.	192.168.50.254
AiMesh nodes (optional)	EMCOMM-NET coverage extension · seamless roaming. Recommended: additional ASUS RT-BE58 Go units.	DHCP assigned
UniFi Switch Lite 16 PoE (recommended)	Wired backbone · powers PoE devices · connects all Pi units, router, cellular antenna, and workstations	DHCP assigned
Operator workstations	Any device with a modern browser: smartphones, tablets, laptops, Raspberry Pi 500 desktops, Windows or macOS laptops	DHCP assigned

- The default server address is 192.168.50.1. This is configurable during installation. All documentation uses this default — substitute your address wherever 192.168.50.1 appears if your deployment differs.

1.4 Capability Summary

CAPABILITY AREA	TOOLS PROVIDED
Incident Management	Five-section ICS structure. Complete IAP form set: ICS-202 through ICS-221. Live T-card resource board. Pre-planned event templates for 6 incident types. Incident archive, restore, and scenario/exercise mode.
ICS Forms & IAP	Full ICS form set with digital signature capture on all prepared-by and approved-by fields. One-click IAP PDF compilation with title page and section dividers. Print center for on-site IAP packages.
FEMA Documentation	FEMA PA cost tracking: Force Account Labor with fringe, Equipment with 2025 FEMA rate schedule (44 built-in rates), Materials, and Contracts. Real-time cost dashboard. ICS-214 import. Project Worksheet text export.

CAPABILITY AREA	TOOLS PROVIDED
Amateur Radio EMCOMM	Net control logger with FCC callsign auto-fill from offline 800,000+ licensee database. APRS tactical map. Winlink and Pat radio email. JS8Call HF digital. AMPRNet 44Net gateway. NTS radiogram generator.
Public Safety Comms	Trunked/P25 net logger with radio ID check-in. EMA member ID lookup. ICS-309 export. Observer mode for read-only net monitoring on any device.
Situational Awareness	Animated NEXRAD radar (WAN required). Live NWS weather alerts. GPS-tracked resource map with status-coded SVG markers. Offline APRS tactical map. SARTopo GeoJSON import. HF propagation tool.
Personnel & Check-In	Member roster with certifications, radio IDs, and QR check-in codes. QR/barcode camera scan check-in using native BarcodeDetector API. ICS-211 check-in log. Digital ID card generator.
WAN & Connectivity	Dual WAN source support: any cellular modem, hotspot, or satellite dish. Preferred/fallback role configuration. Configurable detection per source. Automatic failover. Real-time WAN status with 30-second polling.
Reference & Offline Content	Kiwix offline Wikipedia and reference library. Radio frequency cheat sheets. Hospital and facilities directory. Repeater database. Channel library. NIMS resource typing library. ICS position checklists.

2

Getting Started — Connecting to FieldCommand

<http://192.168.50.1>

Every tool in FieldCommand IMS is a web page served from the Pi. No app installation is required on any device. Any modern browser works — Chrome, Firefox, Safari, Edge, or the built-in browser on any smartphone or tablet.

2.1 First Connection

1. Power on the FieldCommand Pi and wait approximately 45 seconds for all services to start. The system is ready when the EMCOMM-NET Wi-Fi network appears in the device scan list.
2. On any smartphone, tablet, or laptop — open Wi-Fi settings and connect to **EMCOMM-NET**. The password is on the equipment case label and in the Installation Guide. No other credentials are required.
3. Open a browser and navigate to **<http://192.168.50.1>**. The FieldCommand dashboard loads immediately.
4. Bookmark the dashboard URL. On a smartphone, tap **Share** → **Add to Home Screen** to create an icon — it opens like a native app.
5. If the dashboard does not load, confirm you are connected to EMCOMM-NET and not to another Wi-Fi network. The Pi does not relay traffic to the internet — if your device switches to a different network automatically, disable auto-join on other networks during activations.

2.2 Dashboard Modes

The dashboard has three selectable modes that filter the tool card grid. Switching modes does not affect any data — all tools remain accessible in every mode.

MODE	TOOLS SHOWN PROMINENTLY	BEST FOR
All-Hazards ICS	Full ICS platform, IAP forms, T-card resource board, personnel check-in, FEMA cost documentation, event templates, cost dashboard	Any active incident requiring ICS structure
Amateur Radio EMCOMM	Net control logger, FCC callsign lookup, APRS map, Winlink, JS8Call, AMPRNet gateway, HF propagation, NTS radiogram	Amateur radio net operations, ARES/RACES activations
Public Safety	Starcom net logger, radio ID roster, resource map, WAN status, weather radar, hospital directory, facilities	Public safety radio nets, Starcom check-in operations

2.3 WAN Status Indicators

The top of the dashboard always shows a live WAN status bar. It polls every 30 seconds and changes color automatically.

INDICATOR	MEANING	CORE TOOLS AFFECTED?
■ Green — Cellular	Primary internet source active	None — fully operational

INDICATOR	MEANING	CORE TOOLS AFFECTED?
■ Blue — Satellite	Satellite failover active	None — fully operational
■ Red — Offline	No internet — all core tools continue	None — fully operational
■ AMPRNet UP	44Net WireGuard tunnel active	N/A
■ AMPRNet DOWN	44Net tunnel down or gateway Pi unreachable	N/A

A red WAN indicator means internet-dependent features are paused — NWS radar, APRS-IS, HF propagation. It does NOT mean FieldCommand IMS is unavailable. All ICS tools, forms, net loggers, roster, and local maps run normally offline.

3

Organization Setup & Station Configuration

<http://192.168.50.1/setup.html>

Before using FieldCommand IMS for the first time, complete the Organization Setup to configure your station identity, SSID, and default settings. These values appear in ICS form headers, net logs, and printed documents.

3.1 Setup Fields

FIELD	WHAT IT CONFIGURES
Organization name	Full name of your agency or group — appears in ICS form headers and cover pages
Short name / abbreviation	Used in compact displays, footer lines, and net log headers
Primary callsign	Club or personal FCC callsign — used for AMPRNet authentication and net logs
Station grid square	Maidenhead grid locator (4 or 6 character) — used for HF propagation calculations and APRS position reports
Default incident name	Pre-fills the incident name field when creating new incidents
Wi-Fi network name (SSID)	Default is EMCOMM-NET — change if you operate multiple FieldCommand systems simultaneously and need to distinguish them
Server address	Default 192.168.50.1 — change only if this conflicts with your existing network
Time zone	Used for all timestamps in logs and forms — set to your deployment time zone

3.2 Offline vs. Online Feature Table

FEATURE	OFFLINE ?	NOTES WHEN OFFLINE
ICS platform — all five sections	✓ Full	No change
All ICS forms and IAP	✓ Full	No change
T-card resource board	✓ Full	No change
FEMA cost tracking	✓ Full	No change
Event templates	✓ Full	No change
IAP PDF compilation	✓ Full	No change
Digital signatures on forms	✓ Full	No change
Net control loggers (both)	✓ Full	No change
FCC callsign lookup	✓ Full	Uses local SQLite database — no internet needed
Member roster and QR check-in	✓ Full	No change
GPS resource map	✓ Full	No change
Barcode / QR scan check-in	✓ Full	No change

FEATURE	OFFLINE ?	NOTES WHEN OFFLINE
Offline tactical APRS map	✓ Full	Map tiles local — RF APRS still works
Kiwix reference library	✓ Full	No change
NWS weather alerts	■ WAN	Paused — last alert cached and displayed
NEXRAD animated radar	■ WAN	Tiles unavailable — offline banner shown
APRS-IS internet feed	■ WAN	RF APRS continues — internet feed pauses
HF propagation data	■ WAN	Last retrieved data shown until WAN returns
AMPRNet / 44Net tunnel	■ WAN	Tunnel drops — local network unaffected
Winlink Telnet sessions	■ WAN	RF Winlink (VARA/Pactor) continues

4

Member Roster & QR Check-In Codes <http://192.168.50.1/roster.html>

The Member Roster is the central personnel database. It stores every member, mutual-aid visitor, and regular participant with their identifiers, certifications, equipment capabilities, and a unique barcode check-in code that enables rapid QR-scan check-in at any activation.

4.1 Roster Fields

FIELD	PURPOSE
Member ID	Your organization's internal identifier (e.g. ESV-042). Primary key.
Barcode ID	QR/barcode check-in code — defaults to Member ID on first import. Can be overridden with a facility badge number or any custom code.
Callsign	FCC amateur radio callsign. Leave blank for non-licensed members.
Radio ID	Trunked/P25 radio unit number if applicable.
Name	First and last name.
Role	Primary function: Operator, Net Control, Logistics, Medical, etc.
Member type	Member · Visitor · Mutual Aid — affects check-in form behavior.
Certifications	ICS-100/200/300/400/700/800, EmComm I/II, CPR, First Aid, CERT.
Equipment	HF, VHF, digital modes, APRS, Winlink, go-box, generator, vehicle.
License class	Technician, General, Amateur Extra — displayed on net log entries.
Phone / Email	Contact information — stored locally, never transmitted externally.
Grid square	Maidenhead grid locator for RF planning.

4.2 Adding and Editing Members

1. Click **+ Add Member** at the top of the roster page.
2. Enter at minimum a Member ID and name. Add callsign, radio ID, or both if applicable.
3. Check all applicable certifications and equipment capabilities.
4. Click **Save**. The member appears immediately in the roster.
5. To edit an existing member, click their card to open the detail view, make changes, and click **Save**.
6. To delete a member, open their detail view and click **Delete**. This removes them from the roster but does not delete historical net log entries.

4.3 Importing Members via CSV

Import your existing member list using **■ Import CSV**. The CSV must have a **member_id** column. All other columns are optional. Existing member IDs are updated; new IDs are added. Existing members not in the import file are left unchanged.

Required: **member_id**

Optional: `callsign` · `radio_id` · `first_name` · `last_name` · `role` · `phone` · `email` · `grid` · `license_class`

4.4 QR Check-In Codes

Every member has a personal QR code that enables instant check-in at any activation using the Scan Check-In page (see Chapter 10). The QR button on each roster card opens a modal with that member's personal code.

1. Click the **QR** button on any member card.
2. The modal shows the QR code. When internet is available, a true scannable QR image is generated. When offline, the member ID is displayed in large text as a manual entry fallback.
3. Click **Print** to open a clean print-ready page with the QR, the member's name, and the FieldCommand check-in URL.
4. Click **Save PNG** to download the QR code image — members can save it to their phone's home screen or print it on their ID badge.

■ The `barcode_id` field defaults to the `member_id` on first import. If your organization uses physical facility badge scanners, set `barcode_id` to match the badge number — the scan check-in will then auto-fill from a badge swipe.

5

Incident Management — Creating and Managing Incidents <http://192.168.50.1/incident.html>

All ICS work in FieldCommand IMS is organized around an **incident**. An incident holds the ICS forms, resource T-cards, personnel check-ins, cost tracking, and net log associations for one activation or exercise. You can have multiple incidents in the system simultaneously — only one is designated the **active incident** at any time.

5.1 Creating an Incident

1. Navigate to **Incident Management** from the dashboard.
2. Click **+ New Incident**.
3. Enter the incident name, incident number (if assigned by your agency), incident type, and initial operational period.
4. Select the incident commander from your roster, or type a name if the IC is not in the roster. For incidents operating under **Unified Command**, see Section 5.3 below.
5. Click **Create Incident**. The incident opens immediately and becomes the active incident. The incident name appears in the dashboard header and in all form headers.

5.2 Single Incident Commander vs. Unified Command

ICS supports two command structures, and FieldCommand IMS accommodates both without any configuration change. The difference is a workflow convention, not a software setting.

STRUCTURE	WHEN USED	HOW TO CONFIGURE IN FIELDCOMMAND
Single Incident Commander (IC)	One agency or jurisdiction has clear authority. Most activations involving a single responding organization.	Enter the IC's name or callsign in the Incident Commander field when creating the incident. The ICS-203 org chart will show one IC at the top of the command structure.
Unified Command (UC)	Two or more agencies or jurisdictions share command authority — common in multi-agency responses, complex disasters, or incidents crossing jurisdictional boundaries. Each contributing agency provides an IC-equivalent to the UC group. All agencies still work from a single IAP.	Enter "Unified Command" or the UC group designation in the Incident Commander field. List each UC member in the ICS-203 Organization Assignment List under the Unified Command section. All ICS forms, T-cards, and cost documentation continue to function identically — the UC structure is a labeling and documentation convention within the same incident record.

Under Unified Command, each participating agency still develops its own objectives, but those objectives are reconciled into a single IAP. The Planning Section Chief role in the ICS-203 is typically filled by one agency on a rotating basis or by consensus. FieldCommand IMS supports multiple ICS-204 Assignment Lists — one per branch or division — which accommodates multi-agency operational structures naturally.

TYPE	DESCRIPTION
All-Hazards	General ICS incident — suitable for any event not covered below
Search and Rescue	SAR-specific resource typing and form defaults
Severe Weather	Shelter operations, damage assessment, utility coordination

TYPE	DESCRIPTION
Mass Gathering	Large planned event with medical and crowd management focus
HazMat	Hazardous materials response with decon and zoning emphasis
Exercise	Training scenario — marked with ■ badge; beta reset available

5.4 Operational Periods

ICS organizes work into **operational periods** — typically 12 or 24-hour blocks. FieldCommand IMS tracks the current operational period and uses it as the default for new T-cards, check-ins, and form entries. To advance to a new operational period, click **+ New Period** in the incident header. Prior period data is retained and viewable.

5.5 Archive, Restore, and Delete

After an incident closes, the full data package can be archived to a USB drive for long-term storage and removed from the Pi's active storage.

ACTION	WHAT IT DOES	REVERSIBLE ?
Archive to USB	Writes a complete JSON package (all forms, T-cards, check-ins, costs, net log associations) to /media/fieldcommand/backup/incidents/ on the labelled USB drive. Marks the incident as archived on the Pi.	✓ Yes — restore any time
Restore from USB	Reads the JSON archive from the USB drive and re-inserts all data. Restores the incident to fully active status.	✓ Yes
Hard delete from Pi	Permanently removes the incident and all associated data from the Pi SSD. Run only after confirming the archive is on USB.	✗ Permanent

■ The USB drive must be labelled FIELDCOMMAND (all caps). Any USB drive with this label is recognized automatically. A LaCie Rugged or similar ruggedized 1TB USB-C drive is recommended for field use.

5.6 Exercise / Scenario Mode

Exercises and training scenarios can be tagged as ■ **Scenario** when the incident is created. Scenario incidents display a yellow badge throughout the interface so operators always know they are in training mode. A **Beta Reset** wipes all scenario data — incidents, forms, costs, check-ins, T-cards, and meetings — while preserving the roster, hospital directory, channel library, and repeater database. This returns the system to a clean state for the next exercise without losing any permanent configuration data.

6

Pre-Planned Event Templates http://192.168.50.1/event_templates.html

Event Templates allow you to pre-configure a complete incident setup — including objectives, default resource types, communications channels, and ICS organization — and activate it in seconds at the start of an activation. Templates eliminate the repetitive setup work for recurring incident types and ensure consistency across activations.

6.1 Built-In Templates

FieldCommand IMS ships with six built-in templates covering the most common incident types. Each can be used as-is or edited to match your local protocols.

TEMPLATE	PRE-CONFIGURED CONTENTS
Emergency Shelter Operations	Shelter management objectives, cot and supply resource types, registration and medical channels, Red Cross coordination section
Search and Rescue	SAR objectives, field team and K9 resource types, search sector channels, base camp and medical branches
Severe Weather Response	Damage assessment objectives, utility and debris resource types, shelter and EOC coordination channels, public information branch
Mass Gathering / Special Event	Crowd management objectives, medical and security resource types, venue and dispatch channels, medical and operations branches
HazMat Response	Decon and zoning objectives, HazMat team resource types, hot/warm/cold zone channels, safety officer emphasis
Exercise / Training Scenario	Training objectives, evaluator and observer resource types, exercise control channel, scenario-tagged (■)

6.2 Activating a Template

1. Navigate to **Event Templates** from the dashboard.
2. Click the **Activate** tab. Browse the template gallery.
3. Click the template you want. A configuration panel opens.
4. Enter the incident name, date/time, and any other details you want to customize before activating.
5. Click **Activate Template**. FieldCommand IMS creates a new incident pre-loaded with all objectives, resource types, channels, and ICS org structure from the template. You are taken directly to the new incident.

6.3 Creating and Editing Templates

1. Click the **Manage Templates** tab.
2. To edit a template, click its card. The edit modal opens with four tabs: **Objectives**, **Resources**, **Channels**, and **Organization**.
3. Edit any field live. Objectives and resource types are listed — click **+ Add** to add items, click the **×** to remove them.
4. Click **Save Template** when finished.
5. To export a template as a JSON file for sharing with other FieldCommand systems, click **Export JSON**. To import a JSON template, click **Import JSON** and select the file.

6. Built-in templates can be edited but not deleted. Custom templates can be deleted with the **Delete** button.

7

Amateur Radio Net Control Logger <http://192.168.50.1/netcontrol.html>

The Amateur Net Control Logger provides complete net management for licensed amateur radio operations: ARES, RACES, AUXCOMM, and any other amateur net. It replaces paper net logs with a live digital record that feeds directly into the ICS-309 communications log.

7.1 Opening a Net

1. Navigate to **Net Control Logger** from the dashboard.
2. Enter the net name, frequency, and mode (SSB / FM / Digital / Other).
3. Verify the date and time are correct — these become the net open timestamp.
4. Click **Open Net**. The status banner turns green and the elapsed timer starts.
5. Share the  **Observer Link** with section chiefs or served agencies so they can monitor the net in read-only view on their own devices. The observer page auto-refreshes every 15 seconds.

7.2 Logging Check-Ins

When a station checks in, type their callsign in the callsign field. FieldCommand IMS looks up the callsign instantly in the **offline FCC database** — 800,000+ amateur licensees stored locally on the Pi — and fills the operator's name and license class automatically. If the station is on your roster, their member ID is filled as well. No internet required.

1. Type the station's callsign. Name fills automatically as you type.
2. Enter location or remarks if desired.
3. Click **Check In** or press Enter. The entry appears with a timestamp.
4. To check a station out, click **Check Out** on their row. Checkout time and participation duration are recorded automatically, rounded up to the nearest quarter-hour for reimbursement documentation.
5. For traffic, use the traffic entry at the bottom — enter precedence, addressee, and a brief description.

7.3 Closing a Net and Exporting ICS-309

1. Click **Close Net**. All remaining checked-in stations are automatically checked out with the net close timestamp.
2. Net open time, close time, and total duration appear in the header.
3. Click  **Export ICS-309** to download the complete ICS-309 Communications Log as a formatted document ready for the IAP.
4. The ICS-309 includes: net name, frequency, mode, open/close times, total duration, and a full check-in table with individual participation durations rounded up to the nearest quarter-hour.

 The Dead Man's Switch (deadmans.html) monitors net activity and sounds an alert if no check-in is logged within a configurable time window. It is designed for safety monitoring during SAR and field operations where radio silence may indicate an emergency. See Chapter 12.

8

Public Safety Net Logger <http://192.168.50.1/starcom.html>

The Public Safety Net Logger handles trunked radio, P25, and other public safety radio systems where check-in is by radio ID rather than callsign. It is designed for interoperability exercises, served-agency support, and any net where participants may not hold amateur licenses.

FEATURE	AMATEUR NET LOGGER	PUBLIC SAFETY NET LOGGER
Primary ID	FCC callsign — auto-filled from local database	Radio ID (unit number)
License check	FCC ULS lookup confirms active license	None required
Roster lookup	By callsign → name and member ID	By radio ID → name and member ID
ICS-309 format	Callsign in station column	Radio ID in station column
Typical use	ARES / RACES / AUXCOMM	Public safety agency nets, interop exercises

Operation follows the same workflow as the Amateur Net Logger — open the net, log check-ins by radio ID, and export ICS-309 at close. Observer Mode is also available: click **Observer Link** to share a read-only, auto-refreshing view of the active net with any device on EMCOMM-NET.

- Participants who hold both an amateur license and a public safety radio ID should check into the public safety net by radio ID only. If they need to participate in a concurrent amateur net on the same incident, they check into the Amateur Net Logger separately by callsign. This maintains the integrity of both logs.

9

Observer Mode — Read-Only Net View <http://192.168.50.1/observer.html>

Observer Mode provides a read-only, auto-refreshing view of any active net. It is designed for section chiefs, served agency liaisons, EOC duty officers, or anyone who needs to monitor net activity without access to the Net Control workstation.

9.1 Accessing Observer Mode

1. The Net Control operator clicks ■ **Observer Link** on the active net page.
2. The URL is copied to the clipboard. Share it via Winlink, JS8Call, in person, or announce it over the net.
3. Any device on EMCOMM-NET opens the link in a browser. No login required.
4. The observer page shows: net name, frequency, mode, elapsed time, current check-in list with callsigns/names/times, and the traffic log. It refreshes automatically every 15 seconds.

9.2 What Observers Can and Cannot Do

OBSERVERS CAN	OBSERVERS CANNOT
View all check-ins in real time	Add or remove check-ins
View net open time and elapsed duration	Close the net
View the traffic log	Add traffic entries
Refresh the page to get latest data	Change any net settings
Bookmark the observer URL for re-use	Access the Net Control view

10

Barcode & QR Scan Check-In http://192.168.50.1/scan_checkin.html

The Scan Check-In page enables rapid personnel check-in at activations using a smartphone or tablet camera. It uses the browser's native BarcodeDetector API — no external library, no CDN dependency, works fully offline. It supports QR codes, Code 128, Code 39, EAN-13, Data Matrix, PDF417, and Aztec formats.

10.1 Two Check-In Methods

METHOD	HOW IT WORKS	BEST FOR
Camera scan	Point the device camera at the member's QR code or barcode. Detection is automatic at 4 frames per second. The same code is debounced for 3 seconds to prevent duplicate submissions.	High-throughput check-in stations; members with printed QR codes
Manual ID entry	Type a member ID, callsign, or radio ID in the manual entry field and press Enter or click Look Up. Same roster lookup as camera scan.	When camera unavailable; walk-in personnel without a QR code

10.2 Check-In Workflow

1. Navigate to **Scan Check-In** from the dashboard.
2. Click **Start Camera**. If multiple cameras are present (front/rear), select the rear-facing camera from the dropdown.
3. Point the camera at the member's QR code. The system detects it automatically and looks up the member in the roster.
4. If the member is found: their name, agency, and suggested ICS position auto-fill with green borders. Review the fields — edit if needed.
5. If the member is not in the roster: a "not found" banner appears. Their scanned ID fills the ID field. Fill in name and agency manually.
6. Click **Check In**. A full-screen success confirmation appears with the member's name and check-in time. The device vibrates briefly on phones that support it.
7. Click **Scan Next Person** to clear the form and check in the next member. The session history panel at the bottom shows the last 10 check-ins.

10.3 Roster Lookup Order

When a code is scanned or a manual ID is entered, the system searches the roster in this order until a match is found:

SEARCH ORDER	FIELD SEARCHED	EXAMPLE VALUE
1st	barcode_id	ESV-042 (or badge number if overridden)
2nd	member_id	ESV-042
3rd	callsign	KE4CON
4th	radio_id	412



BarcodeDetector is supported on Chrome (Android and desktop), Edge, and Samsung Internet. It is not supported on iOS Safari or Firefox. On unsupported browsers, the camera scan button is disabled and the manual entry fallback works normally.

11

FCC Callsign Lookup <http://192.168.50.1/callsign.html>

The FCC Callsign Lookup tool provides instant offline access to the full national FCC amateur radio licensee database — over 800,000 active licensees. The database is stored locally in SQLite on the Pi and is used automatically by the Net Control Logger, Scan Check-In, and IAP forms. No internet connection is required.

11.1 Manual Lookup

1. Navigate to **Callsign Lookup** from the dashboard.
2. Type any callsign in the search field. Results appear as you type.
3. The result shows: callsign, name, license class, expiration date, grid square, and mailing address city/state.
4. Click **Copy** to copy the callsign to the clipboard, or **Add to Roster** to create a roster entry from the FCC record.

11.2 Automatic Lookup in Other Tools

The callsign database is queried automatically in three other places:

WHERE	HOW IT WORKS
Net Control Logger	As you type a callsign in the check-in field, the name and license class fill automatically. A red border appears if the callsign is not found or the license is expired.
Scan Check-In	After a QR scan or manual entry, if the code is a callsign, the FCC record is used to fill the name field.
ICS-213 / ICS-214	The "From" callsign field in the general message form validates against the FCC database and fills the operator name.

- The FCC database is updated quarterly. To update, download the FCC ULS database export from wireless.fcc.gov/uls and run the database import script as described in the Installation Guide.

12

Dead Man's Switch <http://192.168.50.1/deadmans.html>

The Dead Man's Switch monitors net activity and triggers an audible alert if no check-in is logged within a configurable time window. It is designed for field operations — particularly search and rescue — where radio silence beyond a set interval may indicate a field team emergency.

12.1 Configuration

SETTING	DESCRIPTION
Check-in interval	Maximum time allowed between net check-ins before the alert fires. Common values: 10, 15, 30 minutes. Configurable per operation.
Alert type	Audible tone through the browser (requires speaker). Visual flashing banner on the Dead Man's page.
Reset	Any new check-in in the Net Control Logger resets the timer automatically.

12.2 Operating the Dead Man's Switch

1. Open the Dead Man's Switch page on a dedicated monitor or tablet at the Net Control position or EOC duty station.
2. Select the active net from the dropdown.
3. Set the check-in interval and click **Start Monitoring**.
4. The countdown timer displays time remaining until alert.
5. If the timer reaches zero, the audible alert fires and the page flashes red. Acknowledge with **Acknowledge** and investigate.
6. Any check-in logged in the Net Control Logger resets the timer automatically.
7. Click **Stop Monitoring** when the net closes.

13 Tactical APRS Map <http://192.168.50.1/tactical.html>

The Tactical Map provides an offline-capable, Leaflet-based map showing APRS station positions from both RF APRS (always available) and APRS-IS internet feed (when WAN is up). It is the primary situational awareness tool for tracking field teams, vehicles, and mobile resources.

13.1 Map Layers

LAYER	SOURCE	REQUIRES WAN?
Map tiles	OpenStreetMap / CartoDB Dark	Cached locally for offline use
APRS-IS stations	Internet APRS-IS feed	✓ WAN required
RF APRS stations	Local TNC or direwolf connected to Pi	✗ Offline — always available
SARTopo overlays	GeoJSON import from sartopo_import.html	✗ Offline once imported
Weather alert polygons	NWS API	✓ WAN required

13.2 Station Markers

Each APRS station is displayed with its standard APRS symbol. Clicking a marker opens a popup showing: callsign, timestamp of last beacon, speed and course (if moving), altitude, and the raw APRS comment field. Stations are color-coded by age — bright for recent beacons, faded for stations not heard in more than 30 minutes.

13.3 SARTopo GeoJSON Import

Planning data from SARTopo (search sectors, assignments, exclusion zones) can be exported as GeoJSON and imported into the tactical map using the SARTopo Import page ([sartopo_import.html](#)). Once imported, the overlay persists on the map through browser refreshes.

1. In SARTopo, export your map or layer as GeoJSON.
2. Navigate to **SARTopo Import** from the reference section of the dashboard.
3. Click **Choose File** and select the GeoJSON file.
4. Click **Import**. The overlay appears on the tactical map immediately.
5. To remove the overlay, return to SARTopo Import and click **Clear Overlay**.

14

GPS-Tracked Resource Map http://192.168.50.1/resource_map.html

The Resource Map displays the current GPS positions of all ICS resources for the active incident. Each resource is drawn as a color-coded SVG pin whose color reflects its current status. Resources without a GPS position appear as dashed circles in the sidebar so the Operations Section can see what still needs to be positioned.

14.1 Map Controls

CONTROL	FUNCTION
Incident selector	Choose which incident's resources to display. The last selected incident is remembered between sessions.
■ Refresh	Reload all resource positions from the server.
■ My Location	Place a marker at your current device GPS position and zoom to it. Useful for confirming map accuracy against your known position.
Auto-refresh 30s	Enable automatic position refresh every 30 seconds. Use when resources are actively moving and updating their positions.

14.2 Marker Colors by Status

COLOR	STATUS
■ Green	Available
■ Blue	Assigned
■ Light blue	Staging
■ Red	Out of Service or En Route
Dashed circle	No GPS position set yet

14.3 Setting a Resource Position

Click any resource in the sidebar — or click an existing pin on the map — to open the Set Position modal. Three methods are available:

METHOD	HOW TO USE	BEST FOR
Device GPS	Click ■ Use Device GPS . The browser requests your device's GPS position. Accuracy (±meters) is shown.	Field operator at the resource's location using a phone
Click map to place	Click ■ Click Map to Place . The modal closes and the cursor changes to a crosshair. Click anywhere on the map to set the position.	IC or Planning placing resources on a map by known location

METHOD	HOW TO USE	BEST FOR
Manual coordinates	Type decimal latitude and longitude directly into the fields.	Position known from a paper map, GPS receiver, or radio report

Optionally enter a **Location Label** (e.g. "Division Alpha staging area") to display a human-readable description below the coordinates. Click **Save Position**. The pin appears on the map immediately. Click **Clear GPS** to remove the position.

15

ICS Platform Overview — Five-Section Structure

<http://192.168.50.1/incident.html>

FieldCommand IMS implements the full five-section ICS structure defined by NIMS. Each section has its own page with the forms, tools, and resource displays appropriate to that function. All sections work from the same underlying incident database — a resource added in Operations appears immediately in Logistics and Finance.

ICS SECTION	PAGE	KEY TOOLS
Command	incident.html	Incident overview, general info, ICS-201 briefing, objectives, organizational chart, position checklists, meeting scheduler
Operations	resources.html	T-card resource board by type and status, assignment tracking, GPS resource map, ICS-204 assignment lists
Planning	iap.html	IAP assembly, ICS-202 objectives, ICS-203 org assignment, ICS-205 comms plan, ICS-206 medical plan, Planning P cycle
Logistics	ics-form.html	ICS-215A safety analysis, ICS-218 support vehicle, facilities directory, channel library, resource requests
Finance / Admin	fema_costs.html	FEMA PA cost tracking, Force Account Labor, Equipment with rate schedule, Materials, cost dashboard, ICS-214 import

15.1 Command Structure — Single IC or Unified Command

The ICS-203 Organization Assignment List is where the command structure is formally documented. For a Single IC incident, the IC's name appears at the top of the org chart. For a Unified Command incident, the UC member agencies are listed collectively at the Command level. FieldCommand IMS handles both structures identically — the ICS-203, ICS-202, and all downstream forms work the same way regardless of command structure. See Chapter 5.2 for guidance on configuring each structure.

15.2 General Info — ICS-201 Initial Briefing

The General Info page (general_info.html) holds the ICS-201 Initial Incident Briefing data: situation summary, initial response actions, current organization, and resource summary. It is typically completed by the first-arriving IC and updated through the first operational period.

15.3 Preflight Deployment Checklist

The Preflight page (preflight.html) provides a go/no-go checklist covering hardware, software, power, communications, and personnel readiness before deploying the FieldCommand system. Each item can be checked off and notes added. The checklist can be printed for signature.

16

ICS Operations Section — T-Card Resource Board

<http://192.168.50.1/resources.html>

The T-Card Resource Board is the Operations Section's primary resource tracking tool. It mirrors the physical ICS T-card system used in traditional incident management but adds live status, assignment tracking, GPS integration, and direct export to ICS-204 assignment lists.

16.1 Resource Types

Resources are organized by NIMS type. The Resource Types library ([resource_types.html](#)) defines the types available in your system. FieldCommand ships with standard NIMS types pre-loaded: Crew, Engine, Helicopter, Dozer, Water Tender, Medical Unit, Strike Team, Task Force, and Single Resource. Custom types can be added for your specific operations.

16.2 T-Card Fields

FIELD	DESCRIPTION
Resource name	Unit designation (Engine 12, Team Alpha, Unit 412)
Resource type	NIMS type from the resource types library
Status	Available · Assigned · Staging · Out of Service · En Route
Assignment	Current task or division/group assignment
Leader	Supervisor name or callsign
Personnel count	Number of personnel on this resource
Category	Agency, mutual aid source, or contractor
Daily cost / Rate	Used by FEMA cost tracking and cost dashboard
GPS position	Latitude, longitude, location label — set from Resource Map
Operational period	Which OP period this T-card was created for

16.3 Adding and Managing Resources

1. On the T-Card board, click **+ Add Resource**.
2. Fill in the resource name, type, status, and assignment.
3. Click **Save**. The T-card appears on the board.
4. To change status, click the status badge on the T-card and select the new status.
5. To reassign, click the assignment field and type the new assignment.
6. To set or update GPS position, click **■ Set Position** or go to the Resource Map (Chapter 14).
7. To close out a resource, set status to **Out of Service** and fill in the demobilization notes.

17 ICS Planning Section & IAP Assembly <http://192.168.50.1/iap.html>

The Planning Section manages the Incident Action Plan — the written work plan for each operational period. FieldCommand IMS covers the full ICS form set needed to build a complete IAP, with digital signature capture and one-click PDF compilation.

17.1 IAP Form Set

FORM	TITLE	SECTION
ICS-202	Incident Objectives	Planning
ICS-203	Organization Assignment List	Planning
ICS-204	Assignment List (per branch/division)	Operations
ICS-205	Incident Radio Communications Plan	Planning /Comms
ICS-205 A	Communications List	Planning /Comms
ICS-206	Medical Plan	Planning /Medical
ICS-207	Incident Organization Chart	Command
ICS-208	Safety Message / Plan	Command/Safety
ICS-209	Incident Status Summary	Planning
ICS-211	Incident Check-In List	Planning
ICS-213	General Message	Any section
ICS-214	Activity Log	Any section
ICS-215 A	Incident Action Plan Safety Analysis	Safety/Planning
ICS-218	Support Vehicle/Equipment Inventory	Logistics
ICS-221	Demobilization Check-Out	Planning /Demob
ICS-309	Communications Log	Comms Unit

17.2 Digital Signature Capture

All 16 Prepared By and Approved By fields across the ICS form set support digital signature capture. A signature pad appears below each signature field on every form. Signatures are stored as white-background PNG images and are embedded in the printed form and the IAP PDF.

1. Click on any **Prepared By** or **Approved By** signature field.
2. A signature pad opens below the field.
3. Sign using mouse, touchscreen, or stylus. The signature supports pressure sensitivity on compatible styluses.
4. Click **Accept**. A preview of the signature appears below the field.
5. The signature is saved automatically and restored if the page is reloaded.
6. To clear a signature, click **Clear** on the signature pad.

17.3 IAP One-Click PDF Compilation

The IAP Compile page (iap_compile.html) assembles all completed ICS forms for the active incident into a single print-ready PDF with a cover page and section dividers. This is the IAP package distributed to section chiefs at each operational period briefing.

1. Navigate to **IAP Compile** from the dashboard or the Planning section.
2. The page shows a checklist of all ICS forms for the active incident. Completed forms are shown with a checkmark. Incomplete forms are flagged.
3. Select which forms to include using the checkboxes. Typically include: ICS-202, 203, 204, 205, 206, and 207 at minimum.
4. Click **Compile IAP PDF**. The server generates the PDF server-side using ReportLab and pypdf. This takes 5–15 seconds.
5. The PDF downloads automatically. It includes a title page with the incident name, operational period, and compilation timestamp; section dividers; and all selected forms with embedded signatures.
6. Print the PDF at the Print Center (Chapter 27) for distribution.

■ The IAP PDF is generated on the Pi server — not in the browser. It can be generated from any device on EMCOMM-NET without requiring a printer to be attached to that device. Print from any device through the Print Center.

18

FEMA PA Cost Documentation http://192.168.50.1/fema_costs.html

The FEMA PA Cost Documentation module tracks all reimbursable costs for FEMA Public Assistance (PA) grant documentation. It covers the four FEMA cost categories: Force Account Labor, Equipment, Materials, and Contracts. All entries are tied to the active incident and operational period.

18.1 Force Account Labor

Force Account Labor tracks regular-time and overtime hours for your organization's employees and volunteers. For FEMA PA purposes, only overtime hours are reimbursable for regular employees; all hours may be reimbursable for volunteers depending on your state's policies.

FIELD	FEMA REQUIREMENT
Employee name	Full name as on payroll records
Job title / class	Official job classification
Regular hours	Regular-time hours worked on incident
Overtime hours	Overtime hours — primary FEMA reimbursable category
Hourly rate	Regular hourly rate from payroll
Fringe benefit rate	Fringe as percentage of wages (FEMA requires fringe documentation)
Total with fringe	Calculated automatically: $(\text{hours} \times \text{rate}) \times (1 + \text{fringe}\%)$
ICS-214 import	Import hours directly from a completed ICS-214 Activity Log

18.2 Equipment

Equipment entries use the FEMA Schedule of Equipment Rates. FieldCommand IMS includes all 44 standard FEMA equipment categories pre-loaded from the current year schedule. Select the equipment type and the applicable rate fills automatically.

1. Click **+ Add Equipment Entry**.
2. Click **Lookup** to open the rate picker. Search by equipment type — the FEMA rate fills automatically.
3. Enter hours used and the equipment identifier (unit number or VIN).
4. Add the operator name if the operator is separate from the equipment.
5. The total calculates automatically: $\text{hours} \times \text{FEMA rate}$.

18.3 Materials and Contracts

Materials entries document consumable supplies purchased specifically for the incident. Contracts entries document work performed by contractors. Both require: vendor name, description, purchase order or contract number, unit cost, quantity, and total. Attach receipts and PO documentation as noted — FEMA requires source documentation for all cost claims.

18.4 Project Worksheet Export

Click ■ **Export PW Text** to generate a formatted text summary of all cost entries suitable for copying into the FEMA Grants Portal or attaching to a Project Worksheet. The export includes incident name, dates, itemized costs by category, and total claimed amount.

- FEMA equipment rates are updated annually. The system displays a reminder when the loaded rates are more than one year old. Update rates using the FEMA Equipment Rates page (Chapter 19).

19

FEMA Equipment Rate Schedule http://192.168.50.1/fema_rates.html

The FEMA Equipment Rates page maintains the Schedule of Equipment Rates used by the FEMA cost documentation module. FieldCommand IMS ships with all 44 standard FEMA equipment categories pre-loaded from the 2025 schedule. Rates can be updated annually as FEMA publishes new schedules.

19.1 Viewing and Searching Rates

1. Navigate to **FEMA Equipment Rates** from the Finance section.
2. The table shows all 44 categories with FEMA code, description, unit (hour or day), and current rate.
3. Use the search box to filter by equipment type or FEMA code.
4. A **Rate Year** badge shows which year's schedule is loaded. A yellow warning appears when rates are more than one year old.

19.2 Updating Rates

1. Download the current FEMA Schedule of Equipment Rates from fema.gov.
2. On the FEMA Equipment Rates page, click — **Edit** on any row to update that rate. Enter the new rate and click **Save**.
3. After updating all changed rates, click **Update Rate Year** and enter the current year. This clears the stale-rates warning.
4. Rates are stored in the local database and used immediately by the FEMA cost documentation module.

20

Cost Tracking Dashboard http://192.168.50.1/cost_dashboard.html

The Cost Dashboard provides a real-time financial overview of the active incident. It aggregates all FEMA cost entries and T-card daily rates into a summary view updated automatically every two minutes.

20.1 Dashboard Panels

PANEL	WHAT IT SHOWS
Total incident cost	Running total of all documented costs across all categories. Large tile at the top — visible at a glance from across the room.
Cost by category	Horizontal breakdown bars: Labor · Equipment · Materials · Contracts. Shows both dollar amount and percentage of total for each category.
Resources by type	Personnel count and cost contribution from T-card resources, grouped by resource type.
Cost projection	Extrapolated cost based on current daily burn rate × remaining operational periods. Useful for budget management during multi-day incidents.
Budget tracker	Compare actual costs against an authorized budget amount. Enter the budget figure to see remaining balance and percentage used.
Per-period breakdown	Cost summary broken out by operational period — useful for after-action documentation and FEMA PW preparation.

- The cost dashboard draws data from both the FEMA cost entries (fema_costs.html) and the daily rate fields on T-cards. Set daily or hourly rates on each T-card to see resource costs reflected automatically in the dashboard.

21

ICS-213 General Message <http://192.168.50.1/ics213.html>

The ICS-213 General Message is the standard ICS form for written communications between sections, agencies, or individuals during an incident. FieldCommand IMS provides a fillable ICS-213 with digital signature capture, print output, and Winlink integration.

21.1 Completing an ICS-213

1. Navigate to **ICS-213 General Message** from the dashboard.
2. Fill in: To, From (callsign auto-fills name from FCC database), Subject, and Date/Time.
3. Type the message in the message body field.
4. If a reply is required, check **Reply Requested**.
5. Click the **Prepared By** signature field and sign with mouse, touch, or stylus (see Chapter 17.2 for signature instructions).
6. Click **Print** to print the completed form, or **Send via Winlink** to route the message through Winlink if a Winlink session is active.

21.2 Winlink Integration

A completed ICS-213 can be exported directly to Winlink Express or Pat as a formatted Winlink ICS-213 message. The recipient address, subject, and message body are pre-filled. See Chapter 29 for Winlink configuration.

22

ICS-214 Activity Log & ICS-309 Communications Log

<http://192.168.50.1/ics214.html>

The ICS-214 Activity Log records the activities and significant events for each resource or section during an operational period. Completed ICS-214 entries can be exported directly to FEMA PA labor cost documentation, bridging the gap between activity logging and reimbursement paperwork.

22.1 ICS-214 Activity Log

1. Navigate to **ICS-214** from the Planning section.
2. Select the resource or section this log covers.
3. For each significant activity, click **+ Add Entry**: enter the time, activity description, and resources involved.
4. At the end of the operational period, click **Prepared By** and sign the form.
5. Click **Export to FEMA Labor** to push hours from this log directly into the FEMA Force Account Labor documentation for the active incident.

22.2 ICS-309 Communications Log

The ICS-309 is generated automatically from Net Logger data. It can also be filled manually using the ICS-309 form ([ics309.html](#)) for radio traffic not captured in the Net Logger.

1. Navigate to **ICS-309** from the Communications section.
2. Select the net or communications log to reference.
3. Add manual entries for traffic not logged in the Net Logger.
4. Click **Export ICS-309** to download the formatted log for the IAP.

23

WAN Settings & Dual-Source Internet Configuration

http://192.168.50.1/wan_settings.html

FieldCommand IMS supports two simultaneous internet sources — primary and fallback — with automatic failover between them. The WAN Settings page configures how each source is detected and how the system monitors them. No carrier names or hardware brands are hardcoded — the system works with any cellular modem, phone hotspot, satellite dish, or fixed broadband.

23.1 WAN Source Configuration

FIELD	OPTIONS AND DESCRIPTION
Role	Preferred — the primary WAN source. Used first when available. Fallback — used only when the preferred source is unavailable.
Label	Human-readable name shown on the dashboard status bar: "Cellular", "Satellite", "Hotspot", etc.
Type	Cellular · Satellite · Hotspot · Fixed broadband · Other
Detection method	How the monitor determines if this source is up. See the table below for all methods.
Enabled	Toggle a source on or off without deleting its configuration.

23.2 Detection Methods

METHOD	HOW IT WORKS	BEST FOR
internet_only	Attempts an outbound internet connection. Succeeds if any internet path exists. Does not ping a gateway.	Phone hotspots, USB cellular dongles, any source without a fixed gateway IP
ping	Pings a specific IP address — typically the modem or router gateway. Enter the gateway IP in the field provided.	Sources with a known gateway IP (e.g. a cellular router on 192.168.1.1)
admin_reachable	Checks if the modem's admin web interface is reachable at a given URL. Faster than a full internet check for modems with local admin pages.	Cellular modems with a local admin interface

23.3 Swapping Preferred and Fallback

To swap which source is preferred and which is fallback, click the **Swap Roles** button on the WAN Settings page. This takes effect immediately without restarting any services. Use this when you want to manually prefer satellite over cellular, for example, without changing your physical connections.

■ The WAN monitor runs as a background service and updates the dashboard status bar every 30 seconds. If you change WAN settings, allow up to 60 seconds for the new configuration to take effect.

24 NWS Animated Radar <http://192.168.50.1/radar.html>

The NEXRAD Radar page displays animated radar loops from the National Weather Service using NWS RIDGE II tiles. It requires an active internet connection and shows an offline banner when WAN is unavailable. The radar page is accessible from all three dashboard modes.

24.1 Radar Controls

CONTROL	FUNCTION
■ / ■ Play/Pause	Start or stop the radar animation loop
■ / ■ Step	Advance one frame at a time — useful for analyzing storm movement
Timeline scrubber	Jump to any point in the loaded radar loop
Speed selector	0.5× · 1× · 2× · 4× — controls animation playback speed
Palette	Standard (reflectivity color scale) · Dual-pol · Enhanced
Station selector	Select the nearest NEXRAD radar station for your deployment area
Auto-refresh	Loads new radar data every 5 minutes automatically
Reflectivity legend	Color scale overlay showing dBZ values and associated precip rates

24.2 Fallback When Offline

When the WAN connection is unavailable, the radar page displays an "Offline — radar unavailable" banner and shows the last successfully loaded frame with its timestamp. This allows you to see the most recently received radar image until connectivity is restored. The page polls for WAN restoration every 30 seconds and resumes loading new tiles automatically when internet returns.

25

HF Propagation Tool <http://192.168.50.1/propagation.html>

The HF Propagation tool retrieves current band conditions from the Space Weather Prediction Center (SWPC) and displays a band-by-band propagation forecast. It requires internet access to fetch current data but displays the last retrieved forecast when offline.

BAND	TYPICAL USE IN EMCOMM
160m / 80m	Regional night-time nets, 0–500 mile coverage
40m	Primary EMCOMM HF band — day/night, 500–2000 miles
20m	Long-haul — 2000+ miles, daytime primary
15m / 10m	Long-haul when solar conditions are favorable
6m / 2m	VHF — not covered by HF propagation tool

The tool shows Solar Flux Index (SFI), A-index, K-index, and predicted propagation quality per band. The K-index is the most immediately useful: K 0–2 is excellent; K 3–4 is fair; K 5+ indicates a geomagnetic disturbance that degrades HF propagation.

26

Winlink Radio Email <http://192.168.50.1/winlink-import.html>

FieldCommand IMS integrates with Winlink — the global radio email network used for ICS messaging over RF when internet is unavailable. The integration covers: outbound ICS-213 messages via Winlink, inbound Winlink form import, and Pat (a cross-platform Winlink client) running alongside FieldCommand.

26.1 Winlink Configuration Options

CLIENT	PLATFORM	TRANSPORT
Winlink Express	Windows laptop	VARA HF, VARA FM, Telnet, Pactor
Pat	Raspberry Pi / Linux	VARA FM (via Wine), Telnet, AX.25
RMS Express	Windows laptop	Pactor (requires SCS modem)

26.2 Winlink Form Import

When Winlink messages arrive containing ICS forms (ICS-213, ICS-214, position reports), they can be imported directly into FieldCommand IMS using the Winlink Form Import page.

1. In Winlink Express or Pat, read the incoming message with the ICS form attachment.
2. Save the form as an XML or HTML file.
3. Navigate to **Winlink Form Import** from the dashboard.
4. Click **Choose File** and select the exported form file.
5. Click **Import**. FieldCommand IMS parses the form data and creates the appropriate ICS form entry in the active incident.

27

AMPRNet (44Net) Gateway <http://192.168.50.1/amprgate.html>

The AMPRNet gateway provides a WireGuard-based tunnel into the 44.0.0.0/8 amateur radio IP network when internet is available. It runs on a dedicated second Raspberry Pi (192.168.50.2) — completely isolated from the primary FieldCommand server. If the gateway fails, nothing on the primary server is affected.

27.1 Access Control

The AMPRNet gateway enforces Part 97 access control: only licensed amateur operators with a valid FCC callsign may authenticate. Authentication is validated against the offline FCC database. All access attempts are logged to `/var/log/amprgate-access.log` with callsign, IP, timestamp, and result. Session tokens have an 8-hour TTL.

27.2 Gateway Status Page

The AMPRNet status page (`amprgate.html`) shows: tunnel up/down status, your assigned 44.x.x.x address, connected peers, and the access log. The public read-only port (9000) is accessible from any device on EMCOMM-NET. Tunnel control is restricted to the Pi console (port 9001, localhost only).

- AMPRNet IP allocation requires registration at portal.ampr.org. Request a /29 subnet under your callsign. See the Installation Guide for the complete AMPRNet setup procedure.

28

JS8Call — HF Digital Messaging <http://192.168.50.1/>

JS8Call is a weak-signal HF digital mode designed for keyboard-to-keyboard messaging and store-and-forward relay. It operates on a Windows laptop connected to the incident's HF transceiver (e.g. an IC-7300 with a Digirig or similar USB audio interface) alongside the FieldCommand system on the same EMCOMM-NET network.

28.1 JS8Call for EMCOMM

CAPABILITY	DESCRIPTION
Keyboard messaging	Real-time chat-style messaging between stations — no callsign required to copy
Store and forward	Messages are stored at relay stations and forwarded when the destination is heard
Heartbeat beacons	Automatic periodic beacons announce your station and provide propagation data
Group messaging	Messages addressed to @EMCOMM, @ARES, @RACES reach all monitoring stations
SNR as low as -24dB	Copies signals far weaker than SSB voice or Winlink VARA

28.2 Integration with FieldCommand

JS8Call traffic is monitored on the EMCOMM-NET network. Incoming messages that match ICS message formats can be manually entered into the ICS-213 or the Net Control Logger. A future integration (planned for a subsequent release) will allow direct JS8Call message ingestion into the incident log.

29

NTS Radiogram Generator <http://192.168.50.1/nts.html>

The NTS Radiogram Generator produces properly formatted ARRL National Traffic System radiograms for formal message handling. NTS radiograms are the standard for health-and-welfare and priority traffic during disasters when telephone and internet are unavailable.

29.1 Radiogram Fields

FIELD	DESCRIPTION
Precedence	ROUTINE · WELFARE · PRIORITY · EMERGENCY
Handling	HXG (delivery by mail), HXC (report delivery), others
Station of origin	Originating callsign
Check	Word count of the text — calculated automatically
Place of origin	City/state of the originating station
Time/Date	When the message was filed — filled automatically
To / Phone	Addressee name and phone number for final delivery
Text	Message body — 25 words maximum for efficient traffic handling
Sent by	Callsign of the operator sending this copy

Click **Generate Radiogram** to produce a formatted radiogram ready to transmit. The word count (check) is calculated automatically. Click **Print** to print on a standard traffic form.

30

Repeater Database <http://192.168.50.1/repeaters.html>

The Repeater Database stores VHF/UHF repeater information for your deployment area. It is used by the ICS-205 communications plan, the channel library, and the cheat sheets. Data is stored locally — no internet required.

30.1 Repeater Fields

FIELD	DESCRIPTION
Callsign	Repeater trustee callsign
Output frequency	Receive frequency (what you listen on)
Offset	± MHz, or simplex
CTCSS / DCS	Tone or digital code required to access
Mode	FM · D-STAR · Fusion (C4FM) · P25 · DMR
Coverage	Coverage description — city/county/regional
Notes	Access restrictions, linking, EchoLink node, etc.
Emergency use	Flag as preferred for emergency use — highlighted in the ICS-205

1. Click **+ Add Repeater**.
2. Fill in all applicable fields.
3. Click **Save**. The repeater appears in the database and is immediately available in the ICS-205 channel picker and the cheat sheets.

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Channel Library http://192.168.50.1/channel_library.html

The Channel Library stores pre-configured communications channels for common operations: mutual aid interoperability, public safety talk groups, amateur simplex and repeater frequencies, and satellite channels. Channels from the library can be inserted directly into the ICS-205 Communications Plan with a single click.

Each channel entry stores: channel name, frequency/talk group, mode, CTCSS tone, purpose, and agency assignment. The library is pre-populated with common interoperability channels and can be customized for your jurisdiction's channel plan.

- NIFOG (National Interoperability Field Operations Guide) channels and VTAC channels are included in the default library. Your jurisdiction's specific mutual aid channels should be added during initial system configuration.

32

Hospital Proximity & Facilities Directory <http://192.168.50.1/hospitals.html>

The Hospital Proximity page and Facilities Directory store the locations and contact information for medical facilities, EOCs, staging areas, and other operational facilities used during incidents.

32.1 Hospital Directory

FIELD	DESCRIPTION
Facility name	Official hospital or clinic name
Address	Street address for navigation
Trauma level	Level I · II · III · IV · not designated
Phone	Main switchboard and emergency department numbers
Helipad	Yes / No — for aeromedical considerations
Distance	Calculated from your station grid square (approximate)
Notes	Diversion status, special capabilities, current capacity

32.2 Facilities Directory

The Facilities Directory (facilities.html) stores staging areas, EOCs, shelters, supply depots, and other operational locations. Facilities appear in the ICS-205 and can be linked to T-card assignments. Each facility stores name, address, type, capacity, and operational notes.

33

Reference Tools — Grid, Cheat Sheets, Resources, Print Center

FieldCommand IMS includes a set of reference and utility tools used throughout incident operations. All operate fully offline.

33.1 Grid Square Calculator

The Grid Square Calculator ([grid.html](#)) converts between Maidenhead grid locators and decimal coordinates, and calculates distance and bearing between two grid squares. It supports 4-character (e.g. EN52) and 6-character (e.g. EN52ab) precision.

FUNCTION	INPUT	OUTPUT
Grid to coordinates	Maidenhead grid square (4 or 6 char)	Decimal lat/lon and DMS
Coordinates to grid	Decimal lat/lon	Maidenhead grid square
Distance and bearing	Two grid squares	Distance (km/mi) and true bearing

33.2 Radio Cheat Sheets

The Cheat Sheets page ([cheatsheets.html](#)) provides quick-reference cards for common amateur radio and ICS procedures: phonetic alphabet, Q-codes, RST signal report scale, ICS command structure, NIMS resource typing, and common HF calling frequencies. All cards are printable and designed to fit on a single laminated sheet.

33.3 NIMS Resource Typing Library

The NIMS Resource Typing Library ([resource_types.html](#)) contains the standard NIMS resource type definitions used to populate T-cards. Custom resource types specific to your organization can be added here and immediately become available on the T-card board.

33.4 ICS Position Checklists

The Position Checklists page ([position_checklists.html](#)) provides NIMS-standard activation checklists for each ICS position: IC, Safety Officer, Operations, Planning, Logistics, Finance, and key unit leaders. Each checklist can be checked off during position activation and printed for the position binder.

33.5 Print Center

The Print Center ([printcenter.html](#)) provides optimized print layouts for all ICS forms and documents. From any device on EMCOMM-NET, select the document to print and click Print — the job routes to any printer configured on the Pi via CUPS. Print jobs do not require the operator to be physically at the printer.

34

Network Hardware — Routers, Switch, and Coverage Extension

FieldCommand IMS is designed to work with standard Wi-Fi routers, managed PoE switches, and mesh networking to provide reliable wireless coverage across a deployment site. This chapter covers the recommended hardware configuration and how to extend coverage using AiMesh nodes.

34.1 Recommended Router — ASUS RT-BE58 Go

The ASUS RT-BE58 Go is the recommended primary router for FieldCommand IMS. It is a compact, portable Wi-Fi 7 router designed for mobile deployments — it runs on USB-C power, supports AiMesh for seamless coverage extension, and provides dual WAN ports for simultaneous cellular and satellite sources. It is available from most electronics retailers for approximately \$100–130.

SPEC	RT-BE58 Go
Wi-Fi standard	Wi-Fi 7 (802.11be) — backward compatible with all devices
Bands	2.4 GHz + 5 GHz dual-band
WAN ports	2× — one for cellular/primary, one for satellite/secondary
Power	USB-C — runs from any 65W USB-C PD adapter or battery bank
AiMesh support	✓ — extends EMCOMM-NET coverage with additional RT-BE58 Go units
Typical range	~2,500 sq ft indoors / larger outdoors

34.2 Extending Coverage with AiMesh

AiMesh is ASUS's mesh networking system. Additional ASUS RT-BE58 Go units can be added as AiMesh nodes to extend the EMCOMM-NET signal to additional rooms, floors, or outdoor areas. All nodes use the same SSID and password — devices roam between nodes automatically without reconnecting. Nodes are connected to the primary router via wired Ethernet through the UniFi switch for best performance.

1. Connect the AiMesh node to the UniFi switch with an Ethernet cable.
2. Power on the node. It will appear in the ASUS Router app or the primary router's admin interface at 192.168.50.254.
3. In the router admin interface, navigate to **AiMesh** and click **Add AiMesh Node**. The new node appears automatically.
4. Click **Connect**. The node joins the mesh and begins broadcasting EMCOMM-NET within about 60 seconds.
5. Position nodes to overlap coverage by 20–30% for seamless roaming.

34.3 UniFi Switch Lite 16 PoE

The UniFi Switch Lite 16 PoE is the recommended wired backbone switch. It provides 16 ports with PoE+ on 8 ports — powering PoE devices (cellular antennas, cameras, PoE access points) without separate power adapters. The UniFi management interface provides per-port visibility of link status, speed, and PoE power draw, which is useful for diagnosing connectivity issues in the field.

PORT GROUP	RECOMMENDED USE
Ports 1–2	FieldCommand Pi 5 (primary server) and 44Net Gateway Pi 5
Ports 3–4	ASUS RT-BE58 Go primary router LAN and AiMesh node uplinks
Ports 5–8	PoE — cellular antenna, satellite PoE injector, PoE cameras
Ports 9–16	Operator workstations (Pi 500 units, Windows laptop, tablets)

34.4 Power Considerations

For field deployments without shore power, FieldCommand IMS can run from two Astron RS-35M-AP regulated linear power supplies (one per Pi cluster) fed from a portable generator, or from a high-capacity LiFePO4 battery with a pure-sine inverter. The complete system draws approximately 80–120 watts under typical load. For battery-only operation, a 100Ah 12V LiFePO4 battery provides approximately 8–10 hours of runtime.

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Appendix — Quick Reference & Administration

A.1 All Pages — URL Reference

URL	DESCRIPTION
192.168.50.1/index.html	Main Dashboard
192.168.50.1/incident.html	Incident Management / Command Section
192.168.50.1/incident_mgmt.html	Incident Archive, Restore, Delete, Beta Reset
192.168.50.1/event_templates.html	Pre-Planned Event Templates
192.168.50.1/resources.html	T-Card Resource Board
192.168.50.1/resource_map.html	GPS-Tracked Resource Map
192.168.50.1/resource_types.html	NIMS Resource Typing Library
192.168.50.1/checkin.html	Manual Check-In (ICS-211)
192.168.50.1/scan_checkin.html	QR / Barcode Scan Check-In
192.168.50.1/roster.html	Member Roster and QR Code Generator
192.168.50.1/netcontrol.html	Amateur Radio Net Control Logger
192.168.50.1/starcom.html	Public Safety Net Logger
192.168.50.1/observer.html	Observer Mode — Read-Only Net View
192.168.50.1/deadmans.html	Dead Man's Switch
192.168.50.1/iap.html	IAP Assembly — Planning Section
192.168.50.1/iap_compile.html	IAP One-Click PDF Compilation
192.168.50.1/ics-form.html	ICS Form Suite — all forms
192.168.50.1/ics213.html	ICS-213 General Message
192.168.50.1/ics214.html	ICS-214 Activity Log
192.168.50.1/ics309.html	ICS-309 Communications Log
192.168.50.1/fema_costs.html	FEMA PA Cost Documentation
192.168.50.1/fema_rates.html	FEMA Equipment Rate Schedule
192.168.50.1/cost_dashboard.html	Real-Time Cost Dashboard
192.168.50.1/wan_settings.html	WAN Source Configuration
192.168.50.1/wan-status.html	WAN Status Detail Page
192.168.50.1/radar.html	Animated NEXRAD Radar

URL	DESCRIPTION
192.168.50.1/propagation.html	HF Propagation Data
192.168.50.1/tactical.html	Tactical APRS Map
192.168.50.1/resmap.html	Public Safety Resource Map (Starcom)
192.168.50.1/callsign.html	FCC Callsign Lookup
192.168.50.1/amprgate.html	AMPRNet (44Net) Gateway Status
192.168.50.1/nts.html	NTS Radiogram Generator
192.168.50.1/winlink-import.html	Winlink Form Import
192.168.50.1/briefing_204a.html	ICS-204A Briefing Sheet
192.168.50.1/hospitals.html	Hospital Proximity Directory
192.168.50.1/facilities.html	Facilities Directory
192.168.50.1/repeaters.html	Repeater Database
192.168.50.1/channel_library.html	Channel Library
192.168.50.1/cheatsheets.html	Radio / ICS Cheat Sheets
192.168.50.1/grid.html	Grid Square Calculator
192.168.50.1/position_checklists.html	ICS Position Checklists
192.168.50.1/meetings.html	Meeting Scheduler
192.168.50.1/printcenter.html	Print Center
192.168.50.1/sartopo_import.html	SARTopo GeoJSON Import
192.168.50.1/preflight.html	Preflight Deployment Checklist
192.168.50.1/refs.html	Reference Library
192.168.50.1/setup.html	Organization Setup
192.168.50.1/general_info.html	General Info / ICS-201

A.2 API Ports

PORT	SERVICE	FUNCTION
5050	fcc_lookup_server.py	FCC lookups · incidents · roster · WAN config
5051	ics_platform_server.py	ICS forms · T-cards · check-ins · FEMA costs · GPS
9000	amprgate_status.py	44Net status — public read-only on EMCOMM-NET
9001	amprgate_status.py	Tunnel control — localhost only
80	nginx	Serves all HTML pages and static assets

A.3 Background Services (systemd)

SERVICE FILE	FUNCTION
fieldcommand-main.service	ICS platform server (port 5051)
fieldcommand-fcc.service	FCC lookup and config server (port 5050)
wan-monitor.service	WAN source monitoring and failover
aprs-rf.service	RF APRS receive via direwolf or TNC
amprgate-status.service	44Net status API (ports 9000/9001)
amprgate-poll.service	44Net tunnel keepalive and reconnect
kiwix-serve.service	Kiwix offline reference library server
backup.service / backup.timer	Nightly SQLite backup to USB drive

A.4 Copyright & License

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